

Programming Projects

Introduction



Markus Hohle

University of California, Berkeley

**Numerical Methods for
Computational Science**

MSSE 273, 3 Units



Project 1

Project 2



Project 1

Project 2



Project 1

Objective:

- simulation of pattern formation via diffusion reaction
- finite differences

$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

Fick's 2nd law

c :

concentration

D :

diffusion constant

$$\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}:$$

Laplace operator





Project 1

- Instructions: - see Project_I.ipynb
- Important: - you might see some structures in Project_I.ipynb you don't know yet.
Don't worry, you will learn them soon :)
- Due: - July 25
- Structure: - **Collaborative** Project
→ form teams of 4-5 [here](#) by **July 3rd**
- Deliverables: - each team submits one working code in python (.py)
write names of team members in the code (comments)



Project 1

Project 2



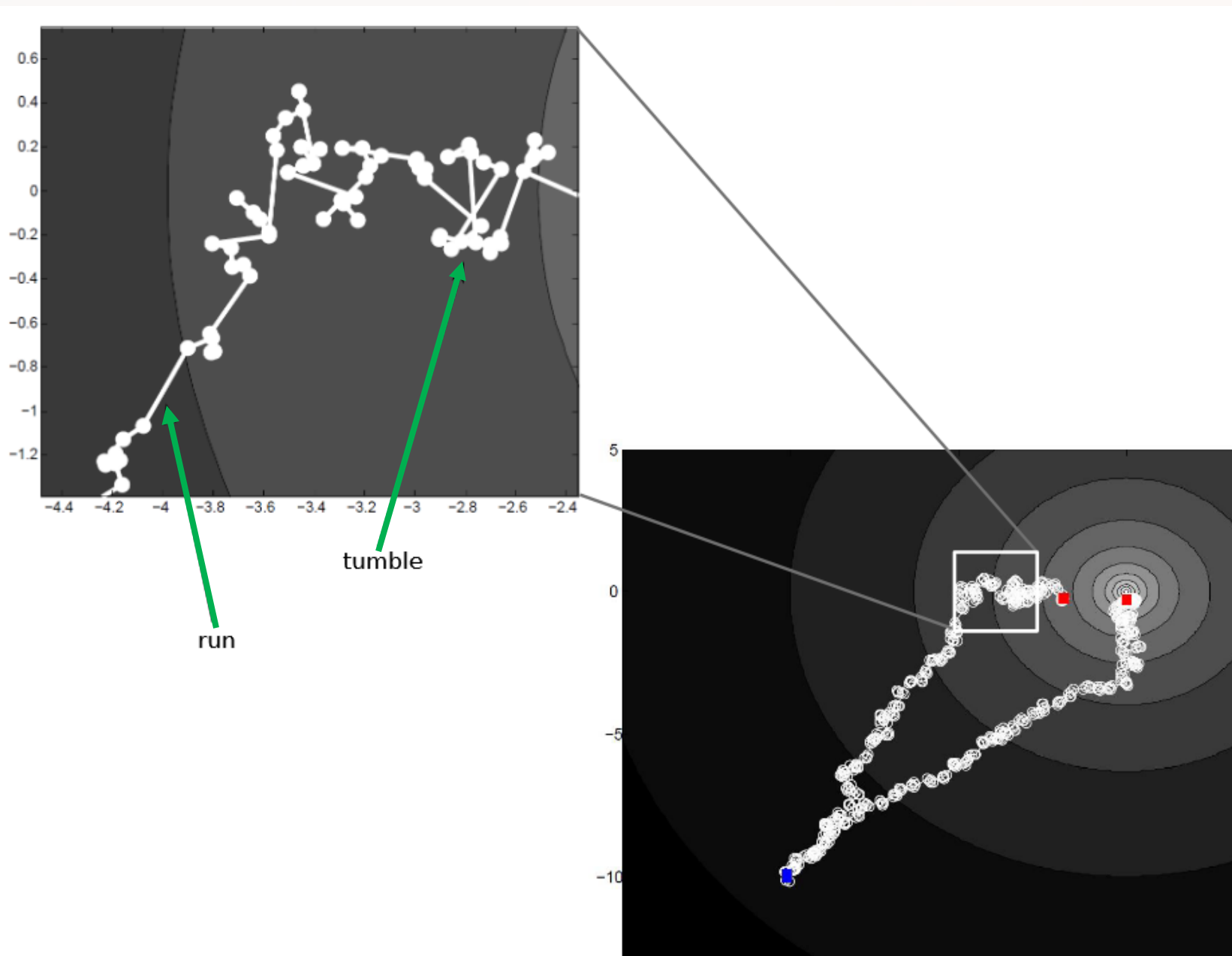
Project 2

Objective

- simulate **biased** random walk of E. Coli
- **Gradient Descent** along a concentration gradient
- $N = 10, 100, 1000 \dots$ E. Coli
- procedure:
 - four subsequent time steps: random walk (**tumble**)
 - calculate concentration gradient from t vs $t - 4\Delta t$
 - move along gradient (**run**)
 - repeat
- plot histogram of locations of E. Coli for different t
- plot motion of E. Coli in $x - y$ plane for different t
- visualize concentration profile
- experiment with three different concentration profiles
- write an actual software package (**functions, modules, classes**)

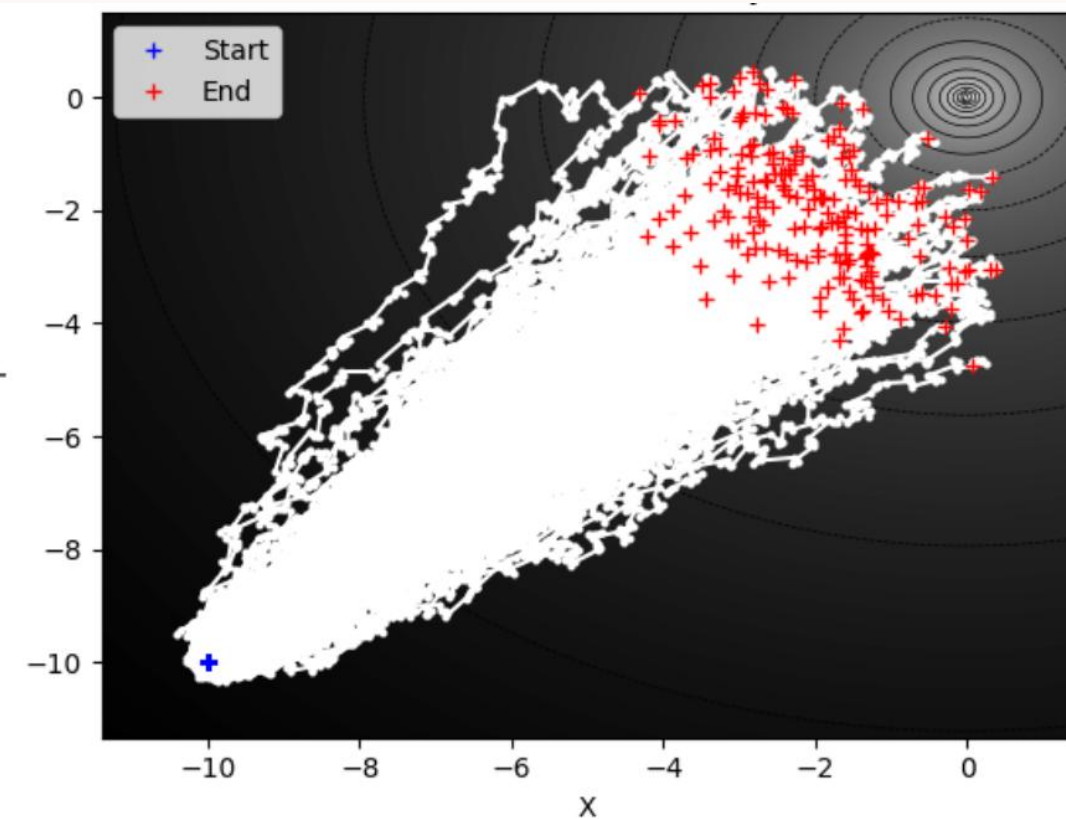
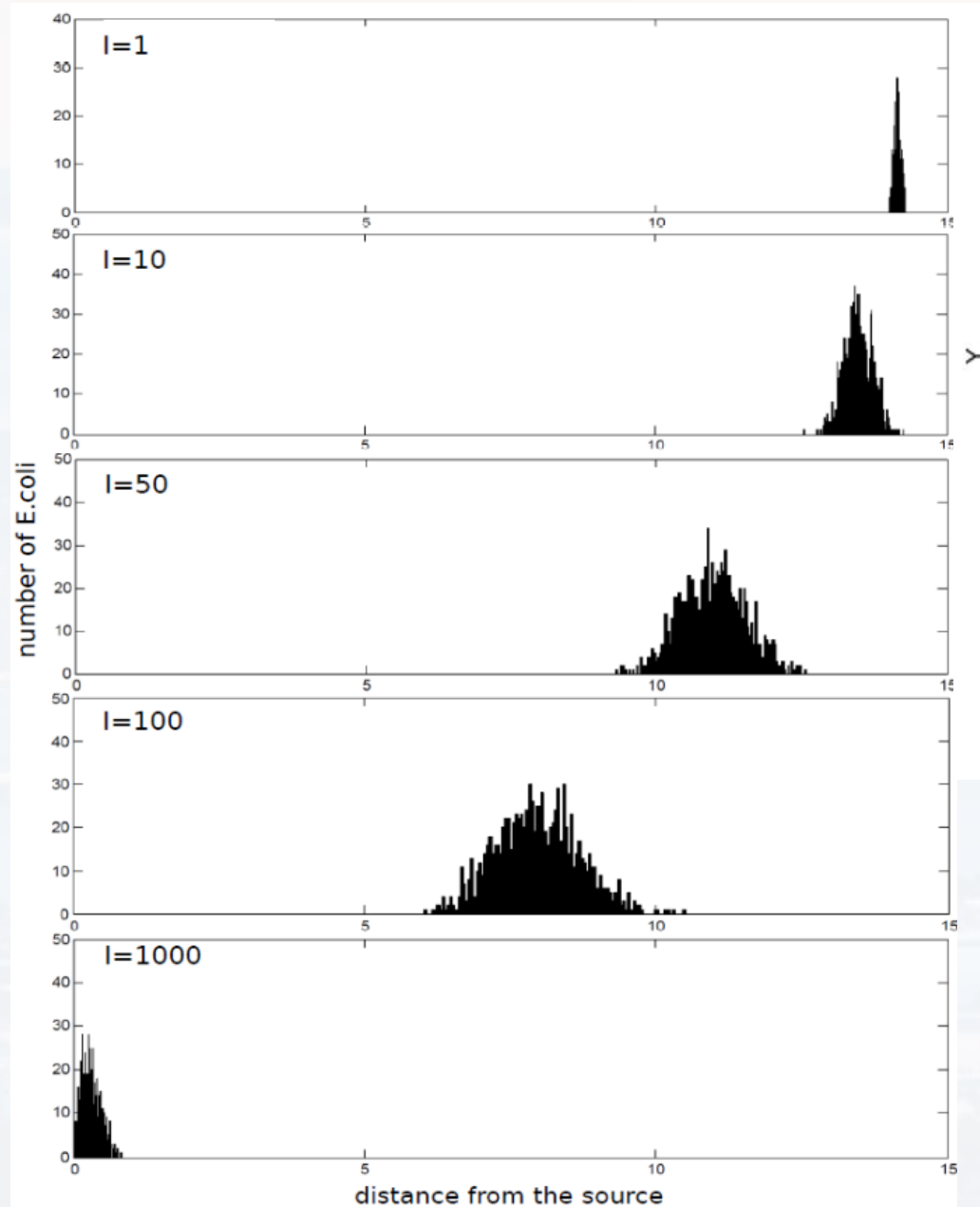


Project 2





Project 2





Project 2

Due: - **Aug 14**

Structure: - **Collaborative** Final Project
→ form teams of 4-5 [here](#) by **July 7th**

Deliverables:

- working code in python (.py)
- a small report (3 – 5 pages)
- use version control (Git/GitHub)

- each team submits **one report** and **one set of code**



Project 2

Report (.pdf):

- Introduction: What is the problem? Why is it important?
- Methods: algorithms, models
- Implementation: How did you do it exactly.
- Evaluation: Present results in a clear and coherent way.
- Discussion: Interpret results, discuss limits of your model and improvements
- References

Code of Honor:

- don't copy/paste someone else's (**including AI**) code
- it should be **your(!)** work
- cite, if you use **any** external source
- not following the code of honor is considered **cheating and will be graded as fail!**



Project 2

SYK Model and its Thermodynamic Properties

Bowen Ouyang, Yaoxing Lu, Zhengqi Wang, Yanzhen Yu, Jie Yang

Abstract

SYK model is a quantum many-body model involving random interactions between fermions. It has a wide range of applications in both condensed matter and high-energy physics. It has a lot of interesting thermodynamic properties that will be discussed later. We will introduce the Hamiltonian, together with a simple recursive algorithm to realize it numerically. We will also talk about the energy spectrum and its properties. Then, we will introduce its thermodynamic properties and discuss their behaviors at low-temperature limit. At last, we will give a brief introduction on supersymmetric SYK model, which is currently a hot topic.

Contents

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1 Introduction

1.1 The Model

The general form of SYK Hamiltonian involves interactions between N Majorana fermions in groups of four:

$$H = i^{\frac{q}{2}} \sum_{1 \leq i_1 < \dots < i_q \leq N} J_{i_1 \dots i_q} \psi_{i_1} \dots \psi_{i_q}$$

Usually, we take N as an even number (denote $N = 2K$), and $J_{i_1 \dots i_q}$ are coupling constants that follow normal distribution with average $\mu = 0$ and standard deviation $\sigma = \sqrt{J(q-1)!/N^{\frac{q-1}{2}}}$ (J is a positive constant).

Good formatting!

Jonathan Cheng, Holden Kowitt, Finnegan Wright

December 13, 2024

Capstone Project Report

Introduction

The aim of this project was to effectively model a fission nuclear reactor using python. The general structure is as follows. Beginning with a set number of neutrons, a Monte Carlo simulation was run to determine the number of fission, scattering, absorption, and escaped neutron events. Determining which events happen utilizes the cross-sections of the different materials in the reactor. A cross-section is a measure of area and can be used to determine the mean free path or the probability of an interaction happening, dependent on the particular elements involved and the energy of a given neutron. Using the cross-sections, cumulative distribution functions can be found which are used in the simulation.

With the number of different events, several interesting physical properties of the reactor can be calculated. Primarily, whether the reactor produces as many neutrons as it consumes; i.e. is the reactor critical. Other results relate to the enrichment of the reactor, the implementation of control rods, manipulating the reflectivity and size of the reactor, as well as observing how many antineutrinos the reactor produces.



Project 2

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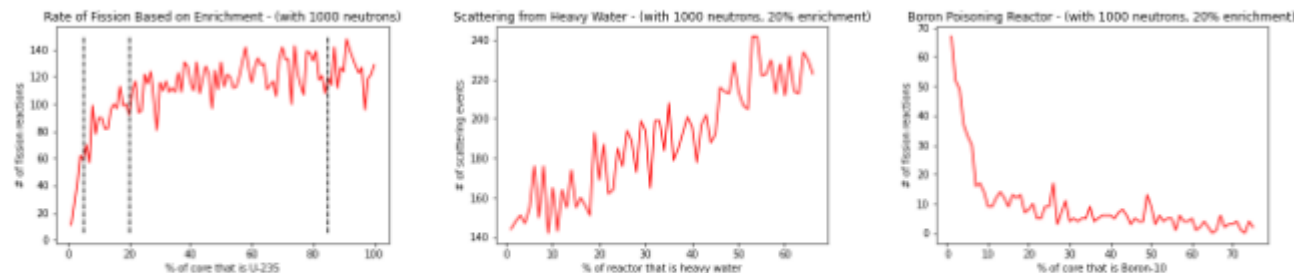


Figure 1: Enrichment, Moderator, Boron Poisoning (left to right)

References

- [1] H. Kim, "Neutron cross section evaluation of u-238." [Online]. Available: <https://t2.lanl.gov/fiesta2017/Talks/Kim.pdf>
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- [7] foxneSsfoxneSs, "Local (?) variable referenced before assignment." [Online]. Available: <https://stackoverflow.com/questions/11904981/local-variable-referenced-before-assignment>
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Project 2

Method of Numerical Integration:

```
40 while (t<20e30):
41     rate(9e20)
42     r12 = star2.pos - star1.pos
43     r13 = planet.pos - star1.pos
44     r23 = planet.pos - star2.pos
45     #positions
46     F12 = -G*m1*m2*norm(r12)/mag(r12)**2 # F12= -F21
47     F13 = -G*m1*m3*norm(r13)/mag(r13)**2 # F13= -F31
48     F23 = -G*m2*m3*norm(r23)/mag(r23)**2 # F23= -F32
49     #forces of the masses (3 bodis)
50     star1.p = star1.p + (-F12-F13)*dt # negative since masses 2 and 3 are pulling on mass 1 (F1 = F31 + F21)
51     star2.p = star2.p + (F12-F23)*dt # only F23 is negative because mass 3 is pulling on mass 2 (F2 = F12 + F31)
52     planet.p = planet.p + (F13+F23)*dt # (F3 = F13 + F23)
53     #momentum
54     star1.pos = star1.pos + star1.p*dt/m1
55     star2.pos = star2.pos + star2.p*dt/m2
56     planet.pos = planet.pos + planet.p*dt/m3
57     #changes the models position
58     t = t+dt
```

Good formatting!

Report

The new positions are the final positions after the system has evolved over time Δt based on the previous iterations' initial position and momentum:

$$r_i(t + \Delta t) = r_i(t) + \frac{p_i(t)}{m_i} \Delta t, \text{ where } \Delta t \text{ is the incremental time step}$$

These new positions are then used to calculate the force experienced by each body using Newton's law of gravity:

$$F_{ij} = \frac{-Gm_i m_j}{|r_{ij}|^2} \hat{r}_{ij}$$



Project 2

Bad formatting!

→ will be graded poorly

→ or even not accepted

are located when the masses and points form an equilateral triangle, i.e., when the angle between each point is 60 degrees. Our derivations of these points are located below.

2-mass system

$$\frac{Gm_1m}{(r_1+r_2)^2} + \frac{Gm_2m}{(r_2-r_1)^2} = \frac{Gm_1m_2}{R^2r_1} + \frac{Gm_2m_2}{R^2r_2}$$

$$\frac{Gm_1}{(r_1+r_2)^2} + \frac{Gm_2}{(r_2-r_1)^2} = \frac{Gm_1}{R^2r_1} + \frac{Gm_2}{R^2r_2}$$

$$\boxed{\frac{m_1}{(r_1+r_2)^2} + \frac{m_2}{(r_2-r_1)^2} = \frac{m_2}{R^2r_1} + \frac{m_1}{R^2r_2}}$$

Lagrange Points (Barycentric star)

$$r_1 = \frac{m_2 R}{m_1 + m_2} \quad r_2 = \frac{m_1 R}{m_1 + m_2}$$

$$v_1 = \sqrt{\frac{Gm_2 r_2}{R^2}} \quad v_2 = \sqrt{\frac{Gm_1 r_1}{R^2}}$$

$$\omega = \frac{v_1}{r_1} = \frac{v_2}{r_2} = \sqrt{\frac{Gm_2}{R^2 r_1}}$$

$$F_{m1} = \frac{Gm_1 m_2}{r_1^2} \quad F_{m2} = \frac{Gm_2 m_1}{r_2^2}$$

$$\frac{Gm_1 m_1}{(r_1+r_2)^2} = \frac{Gm_2 m_1}{(r_2-r_1)^2} = \frac{m_1 Gm_2}{R^2 r_1}$$

$$\boxed{\frac{m_1}{(r_1+r_2)^2} + \frac{m_2}{(r_2-r_1)^2} = \frac{m_2}{R^2 r_1} + \frac{m_1}{R^2 r_2}}$$



Project 2

Please ask Questions During Office Hours and Discussions!